Problems 3

1. Prove that $G \times H$ is abelian if and only if G and H are both abelian.

2. Write each of the following permutations as a product of disjoint cycles.

$$(1\ 2\ 3)(1\ 2) \qquad (7\ 3\ 4\ 5\ 9\ 2\ 6)(6\ 3\ 4)(4\ 3\ 5\ 6\ 7) \qquad (1\ 7)(1\ 6)(1\ 5)(1\ 4)(1\ 3)(1\ 2)$$

$$(1\ 4)(1\ 2\ 3)(4\ 5\ 6\ 7) \qquad (1\ 4)(1\ 2\ 3\ 4\ 5\ 6\ 7)$$

3. Determine the inverse of each of the following elements of S_5.

$$(1\ 2\ 3\ 4) \qquad (1\ 3)(2\ 4) \qquad (1\ 4)(2\ 3\ 5) \qquad (1\ 2)(2\ 3\ 4\ 5)$$

4. Let $g = (1\ 2\ 3\ 4\ 5\ 6\ 7)$. Write down the cycle representation of each of $g^2, g^3, g^4, g^5, g^6, g^7, g^8$.

5. Let $g = (1\ 2\ 3)(4\ 5\ 6\ 7)$. Write down the cycle representation of each of $g^2, g^3, \ldots, g^{13}$.

6. Let $g = (1\ 2)(3\ 4)(5\ 6\ 7)$. Write down the cycle representation of each of $g^2, g^3, \ldots, g^7$.

7. List the elements of S_4, writing each as a product of disjoint cycles.

8. Let G be a group and let $a, b \in G$. Then b is called a *conjugate* of a if and only if there exists $g \in G$ such that $b = gag^{-1}$.

   (a) Prove that conjugacy is an equivalence relation on G. That is, prove that the relation on G where b is related to a if and only if b is a conjugate of a is an equivalence relation.

   (b) Prove that conjugate elements of a group have the same order.

   (c) The equivalence classes of the conjugacy relation are called *conjugacy classes*. Determine the conjugacy classes of each of the three groups $(\mathbb{Z}, +)$, S_3 and D_4.

9. Let X be a set and let $G \leq \mathrm{Sym}(X)$. Define a relation $\sim$ on X by $x \sim y$ if and only if there exists $g \in G$ such that $g(x) = y$.

   (a) Prove that $\sim$ is an equivalence relation on X. (The equivalence classes of $\sim$ are called the *orbits* of X under G.)

   (b) Let X be the set of all 3-element subsets of $\mathbb{Z}_6$. For each $x \in \mathbb{Z}_6$, define a function $\pi_x : X \to X$ by $\pi_x(\{a, b, c\}) = \{a + x, b + x, c + x\}$, and let $G = \{\pi_x : x \in \mathbb{Z}_6\}$. Prove that $G \leq \mathrm{Sym}(X)$ and determine the orbits of X under G.

10. Let S be a finite set, let $G \leq Sym(S)$, let $a \in S$, and let $X \subseteq S$.

    (a) Let $G_a = \{g \in G : g(a) = a\}$. Prove that $G_a \leq G$. (The group G_a is called the *stabilizer of a in G*.)

    (b) Let $G_{\{X\}} = \{g \in G : g(X) = X\}$. Prove that $G_{\{X\}} \leq G$. (The group $G_{\{X\}}$ is called the *setwise stabilizer of X in G*.)

    (c) Let $G_X = \{g \in G : g(x) = x \text{ for each } x \in X\}$. Prove that $G_X \leq G$. (The group G_X is called the *pointwise stabilizer of X in G*.)

    (d) Prove that S_n has a subgroup of order $m!$ for each $m = 1, 2, \ldots, n$.

    (e) Prove that S_n has a subgroup of order $m!(n-m)!$ for each $m = 1, 2, \ldots, n$.

11. (a) Let G be a group and let $a, b \in G$. Prove that if $ab = ba$, then $ab^{-1} = b^{-1}a$.

    (b) Let G be a group, let $a \in G$, and let

    $$C_G(a) = \{g \in G : ag = ga\}.$$

    Prove that $C_G(a) \leq G$. (The group $C_G(a)$ is called the *centralizer of a in G*.)

    (c) Let G be a group and let $Z(G) = \{g \in G : gx = xg \text{ for all } x \in G\}$. Prove that $Z(G) \leq G$. (The group $Z(G)$ is called the *centre of G*.)

12. Draw a diagram showing the symmetries of a regular hexagon. Label each symmetry with the corresponding element of $D_6 = \{1, \sigma, \sigma^2, \ldots, \sigma^5, \tau, \sigma\tau, \sigma^2\tau, \ldots, \sigma^5\tau\}$.

13. Let $n \geq 3$ and let $D_n = \{1, \sigma, \sigma^2, \ldots, \sigma^{n-1}, \tau, \sigma\tau, \sigma^2\tau, \ldots, \sigma^{n-1}\tau\}$ where $\sigma = (1\ 2\ 3\ \cdots\ n{-}1\ n)$, $\tau = (1)(2\ n)(3\ n-1)(4\ n-2)\cdots(\frac{n+1}{2}\ \frac{n+3}{2})$ when n is odd, and $\tau = (1)(2\ n)(3\ n-1)(4\ n-2)\cdots(\frac{n}{2}\ \frac{n+4}{2})(\frac{n+2}{2})$ when n is even so that D_n is the dihedral group of degree n.

    (a) Which listed element of D_n is equal to $\tau\sigma$? Explain.

    (b) Is D_n abelian? Explain.

    (c) Which listed element of D_n is equal to $\tau\sigma^m$? Explain.

    (d) Does D_n have a subgroup of order 2? Explain.

    (e) Prove that D_n has a cyclic subgroup of order n.

14. In each of the following, a group G and a subset S of G is given. In each case give the order of the group $\langle S \rangle$ and list the elements of $\langle S \rangle$.

(a) $G = \mathbb{Z}_{12}$, $S = \{5\}$. (b) $G = \mathbb{Z}_{12}$, $S = \{3\}$. (c) $G = \mathbb{Z}_{24}$, $S = \{4, 6\}$.

(d) $G = \mathbb{Z}_{16}^*$, $S = \{3\}$. (e) $G = \mathbb{Z}_{17}^*$, $S = \{2\}$. (f) $G = D_6$, $S = \{\sigma^2, \tau\}$.

(g) $G = S_7$, $S = \{(1\ 2\ 3), (4\ 5\ 6\ 7)\}$.

(h) $G = S_4$, $S = \{(1\ 2\ 3), (1\ 2)(3\ 4)\}$.